Autodesk® Scaleform®

Scale9Grid

Scaleform Scale9Grid

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Autodesk Scaleform

Scale9Grid User Guide

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1 Flash Scaleform Scale9Grid

Scale9Grid Flash® Scale9Grid

Flash Scale9Grid 9-Slice Scaling

Flash Studio "Symbol properties" 9-Slice Scaling ActionScript MovieClip.scale9Grid Button.scale9Grid Flash Scale9Grid Flash Studio

Adobe Flash Scale9Grid Flash Scale9Grid Scaleform Scale9Grid Flash Scale9Grid Adobe Flash Scaleform

Scale9Grid	**Adobe Flash**	**Scaleform**
	X/Y	
		Scale9Grid
Scale9Grid		

Scaleform Flash Flash Scale9Grid Scaleform Flash Scale9Grid Scaleform

Scale9Grid FLA/SWF Scaleform

2

Flash Scaleform Scale9Grid Flash
Sprite
Flash Scaleform
Scale9Grid

3

Flash Scaleform
Flash
Scale9Grid Scaleform
Scale9Grid
EdgeAA

Scale9Grid
Scale9Grid

Flash
Scale9Grid

Scaleform

Adobe Flash

4

ActionScript

4.1 Flash

Flash
scale0grid_window1.fla scale9grid_window1.swf Flash Studio

Flash ActionScript hit-testing
Flash

ActionScript

```
import flash.geom.Rectangle;

var bounds:Object = this.getBounds(this);
for (var i in bounds) trace(i+" --> "+bounds[i]);
this.XMin = bounds.xMin;
this.YMin = bounds.yMin;
this.XMax = bounds.xMax;
this.YMax = bounds.yMax;
this.MinW = 120;
this.MinH = 100;
this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;
this.XMode = 0;
this.YMode = 0;
```

```
this.onPress = function()
{
	this.OldX = this._x;
	this.OldY = this._y;
	this.OldW = this._width;
	this.OldH = this._height;
	this.OldMouseX = _root._xmouse;
	this.OldMouseY = _root._ymouse;
	this.XMode = 0;
	this.YMode = 0;

	var kx = (this.XMax - this.XMin) / this._width;
	var ky = (this.YMax - this.YMin) / this._height;
	var x1 = this.XMin + 6  * kx;   // Left
	var y1 = this.YMin + 4  * ky;   // Top
	var x2 = this.XMax - 26 * kx;   // Right
	var y2 = this.YMax - 25 * ky;   // Bottom
	var w  = 20 * kx;
	var h  = 20 * ky;
	var xms = this._xmouse;
	var yms = this._ymouse;

	if (xms >= x1 && xms <= x1+w)
	{
		if (yms >= y1 && yms <= y1+h)
		{
			this.XMode = -1;
			this.YMode = -1;
		}
		else
		if (yms >= y2 && yms <= y2+h)
		{
			this.XMode = -1;
			this.YMode =  1;
		}
	}
	else
	if (xms >= x2 && xms <= x2+w)
	{
		if (yms >= y1 && yms <= y1+h)
		{
			this.XMode =  1;
			this.YMode = -1;
		}
		else
		if (yms >= y2 && yms <= y2+h)
		{
			this.XMode =  1;
			this.YMode =  1;
		}
	}

	if (XMode == 0 && YMode == 0)
	{
		this.startDrag();
	}
```

```
}


this.onRelease = function()
{
      this.XMode = 0;
      this.YMode = 0;
      this.stopDrag();
}

this.onReleaseOutside = function()
{
      this.XMode = 0;
      this.YMode = 0;
      this.stopDrag();
}

this.onMouseMove = function()
{
      var dx = _root._xmouse - OldMouseX;
      var dy = _root._ymouse - OldMouseY;

      if (this.XMode == -1)
      {
            this._x     = this.OldX + dx;
            this._width = this.OldW - dx;
            if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
            {
                  this._x     = this.OldX + this.OldW - this.MinW;
                  this._width = this.MinW;
            }
      }
      if (this.XMode == 1)
      {
            this._width = this.OldW + dx;
            if (this._width < this.MinW || _root._xmouse < this.OldX)
                this._width = this.MinW;
      }
      if (this.YMode == -1)
      {
            this._y      = this.OldY + dy;
            this._height = this.OldH - dy;
            if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)
            {
                  this._y      = this.OldY + this.OldH - this.MinH;
                  this._height = this.MinH;
            }
      }
      if (this.YMode == 1)
      {
            this._height = this.OldH + dy;
            if (this._height < this.MinH || _root._ymouse < this.OldY)
                this._height = this.MinH;
      }
}
```

XMode Ymode "-1"
"1"

onPress ( ) kx ky
Scale9Grid

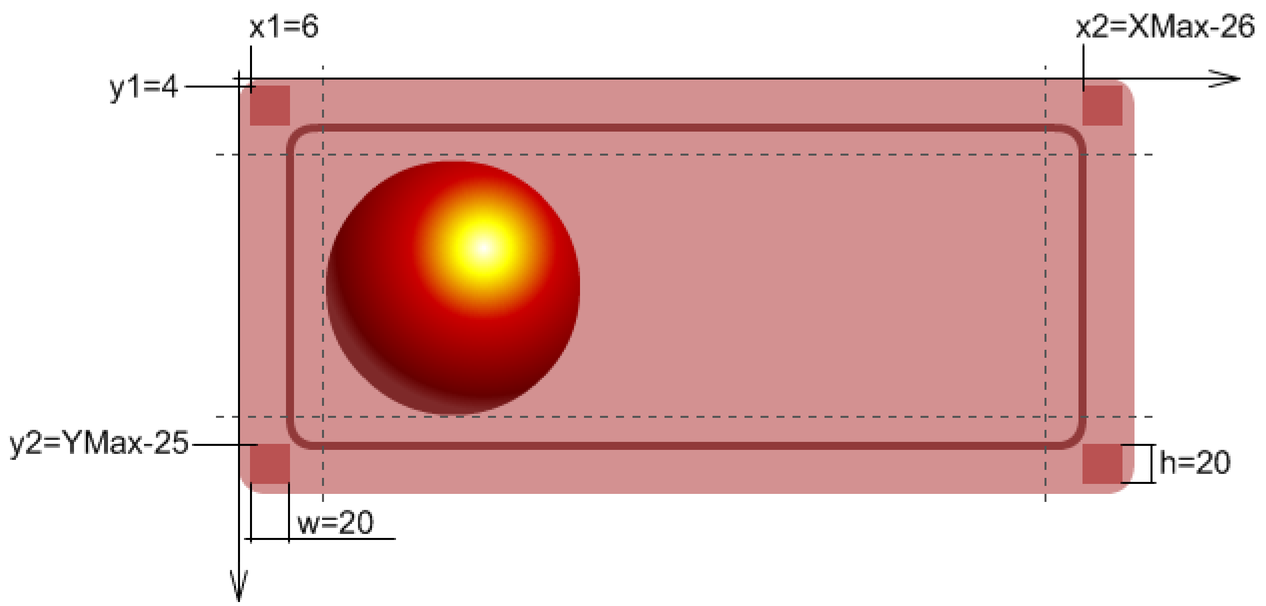
```
var kx = (this.XMax - this.XMin) / this._width;
var ky = (this.YMax - this.YMin) / this._height;
```

Left-Top: x1, y1, x1+w, y1+h
Right-Top: x2, y1, x2+w, y1+h
Left-Bottom: x1, y2, x1+w, y2+h
Right-Bottom: x2, y2, x2+w, y2+h

```
var x1 = this.XMin + 6  * kx;   // Left
var y1 = this.YMin + 4  * ky;   // Top
var x2 = this.XMax - 26 * kx;   // Right
var y2 = this.YMax - 25 * ky;   // Bottom
var w  = 20 * kx;
var h  = 20 * ky;
```

6 4 26 25

ActionScript




```
var xms = this._xmouse;
```

```
var yms = this._ymouse;
if (xms >= x1 && xms <= x1+w)
{
      ... and so on
```

ActionScript

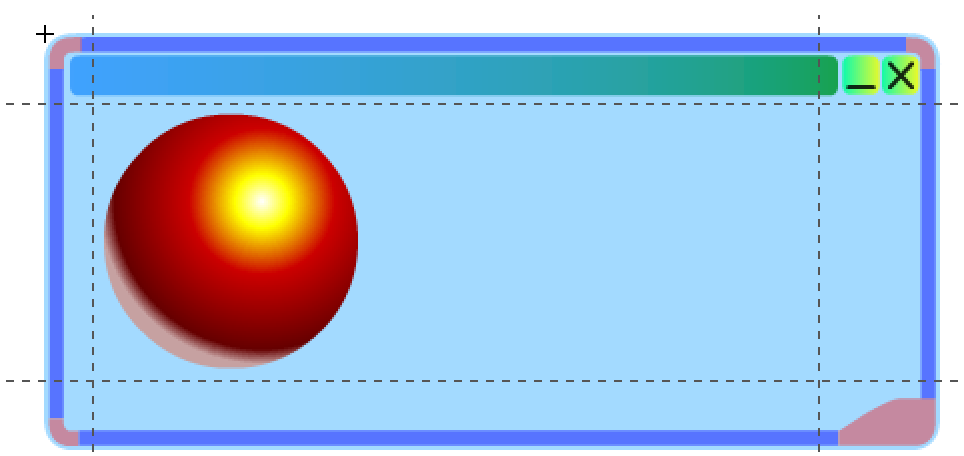
Flash

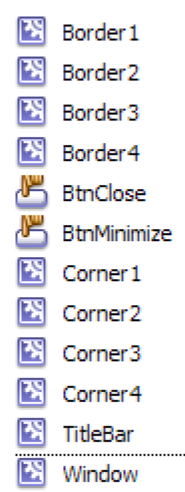
4.2 Scaleform

Scaleform Flash Scaleform

scale9grid_window2.fla scale9grid_window2.swf
ActionScript

“ ”





Flash Border1 Border4 Corner1 Corner4 TitleBar Scaleform Scale9Grid ActionScript OnResize ( )

```
import flash.geom.Rectangle;
//trace(this.scale9Grid);
this.MinW = 100;
this.MinH = 100;
this.XMode = 0;
this.YMode = 0;
this.OldX = this._x;
this.OldY = this._y;
this.OldW = this._width;
this.OldH = this._height;
this.OldMouseX = 0;
this.OldMouseY = 0;

function AssignResizeFunction(mc:MovieClip, xMode:Number, yMode:Number)
{
      mc.onPress          = function() { this._parent.StartResize(xMode, yMode);
      mc.onRelease        = function() { this._parent.StopResize(); }
      mc.onReleaseOutside = function() { this._parent.StopResize();    }
      mc.onMouseMove      = function() { this._parent.OnResize(); }
}

AssignResizeFunction(this.Border1,  0, -1);
AssignResizeFunction(this.Border2,  1,  0);
AssignResizeFunction(this.Border3,  0,  1);
AssignResizeFunction(this.Border4, -1,  0);
AssignResizeFunction(this.Corner1, -1, -1);
AssignResizeFunction(this.Corner2,  1, -1);
AssignResizeFunction(this.Corner3,  1,  1);
AssignResizeFunction(this.Corner4, -1,  1);
this.TitleBar.onPress          = function() { this._parent.startDrag(); }
this.TitleBar.onRelease        = function() { this._parent.stopDrag(); }
this.TitleBar.onReleaseOutside = function() { this._parent.stopDrag(); }

function StartResize(xMode:Number, yMode:Number)
{
      this.XMode = xMode;
      this.YMode = yMode;
      this.OldX = this._x;
      this.OldY = this._y;
      this.OldW = this._width;
      this.OldH = this._height;
      this.OldMouseX = _root._xmouse;
      this.OldMouseY = _root._ymouse;
}

function StopResize()
{
      this.XMode = 0;
      this.YMode = 0;
}
```

```
function OnResize()
{
      var dx = _root._xmouse - OldMouseX;
      var dy = _root._ymouse - OldMouseY;

      if (this.XMode == -1)
      {
            this._x     = this.OldX + dx;
            this._width = this.OldW - dx;
            if (this._width < this.MinW || _root._xmouse > this.OldX + this.OldW)
            {
                  this._x     = this.OldX + this.OldW - this.MinW;
                  this._width = this.MinW;
            }
      }
      if (this.XMode == 1)
      {
            this._width = this.OldW + dx;
            if (this._width < this.MinW || _root._xmouse < this.OldX)
                this._width = this.MinW;
      }
      if (this.YMode == -1)
      {
            this._y      = this.OldY + dy;
            this._height = this.OldH - dy;
            if (this._height < this.MinH || _root._ymouse > this.OldY + this.OldH)
            {
                  this._y      = this.OldY + this.OldH - this.MinH;
                  this._height = this.MinH;
            }
      }
      if (this.YMode == 1)
      {
            this._height = this.OldH + dy;
            if (this._height < this.MinH || _root._ymouse < this.OldY)
                this._height = this.MinH;
      }
}
```

Adobe Flash

ActionScript “ ”

Flash Studio Scale9Grid SWF BUG
scale9grid_window2.fla BUG “TitleBar”
Scale9Grid Flash Studio
Scale9Grid “Save and Compact”

ActionScript Scale9Grid

```
trace(this.scale9Grid);
```

BUG

```
this.scale9Grid = new Rectangle(24, 35, 363, 138);
```

scale9grid_window3.fla scale9grad_window3.swf

Scaleform Flash
Adobe Flash
Scale9Grid Flash
Scaleform Scale9Grid